

Confidence-Aware Edge Routing for Robotic TCM Tongue Imaging: Edge-IQA and LangGraph Closed-Loop Acquisition

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Abstract

Robotic acquisition of tongue images for traditional Chinese medicine (TCM) tele-diagnosis requires tight coupling between motion, on-device quality assessment, and upload policy under variable network delay. We present a cloud–edge–device stack that keeps learning-based perception out of the 1 kHz ROS control loop: a lightweight *Edge-IQA* scores each frame on the gateway, and a *Lang-Graph* router decides among cloud upload, edge resampling, or fail-safe abort within a bounded retry budget. On an offline main experiment (500 frames $\times$ 3 seeds, baselines B0–B3), the proposed B2 configuration reduces median end-to-end RTT (M1 p50) from 397.6 ms to 364.7 ms versus naive upload (B0) while raising valid-frame rate (M2) from 0.50 to 0.99. Edge-IQA runs at p95 $\approx$ 9.6 ms per frame on CPU. An auxiliary Franka closed-loop track (50 trials) confirms routing decisions are reproducible on the execution stack.

1 Introduction

Tele-diagnosis for tongue inspection in TCM needs repeatable, high-quality images from a manipulator-mounted camera before cloud analytics and electronic medical records (EMR) integration [8]. Naive pipelines upload every frame to the cloud, wasting bandwidth on blurred or poorly exposed captures and inflating end-to-end latency when the link is slow.

We study a **cloud–edge–device** architecture for a Franka Research 3 (FR3) platform with ROS 2 and MoveIt motion execution. Perception and routing policies live in a Python edge gateway (`franka_api_server`) and in the `doctor/paper1` research tree—not inside hardware plugins—so algorithms remain testable without a robot.

Contributions.

1. **Edge-IQA:** an $O(HW)$ quality index fusing Laplacian sharpness and exposure, with calibrated threshold $\tau^*=0.505$ and CPU p95 below 10 ms at 512 px.
2. **Confidence-aware LangGraph routing:** a three-branch policy (`upload_cloud`, `resample_edge`, `fail_safe`) with retry budget $K=2$, logged as JSONL for reproducibility.

3. **Dual-track evaluation:** offline baselines B0–B3 supply Table 3 metrics M1–M3; an auxiliary Franka track (M6) validates execution-stack trends without mixing CSVs into primary statistics.

2 Related Work

Robotic medical imaging. Surgical and diagnostic robots increasingly use ROS 2 and MoveIt for motion planning [1, 4]. We adopt named *skills* (calibrated joint poses) instead of online impedance tuning for acquisition poses, matching clinical repeatability requirements.

Image quality assessment. No-reference IQA ranges from hand-crafted sharpness metrics to deep models [5, 7]. Edge deployment favors linear-time operators; our Edge-IQA follows this line with explicit blur/exposure flags.

Edge–cloud orchestration. Split inference and early-exit policies reduce bandwidth in mobile vision [2, 3]. LangGraph-style state machines express branching policies with auditable logs, complementary to monolithic ROS nodes.

TCM tongue analysis. Vision pipelines for tongue classification and color analysis are mature in the cloud [6, 8]. Paper I focuses on *acquisition quality and routing* upstream of these models.

3 Method

4 System Architecture

We target a cloud–edge–device stack for robotic tongue-image acquisition in traditional Chinese medicine (TCM) tele-diagnosis. The design separates *real-time* acquisition and quality control at the edge from *offline* or high-latency cloud analytics, while keeping algorithm modules testable outside the robot operating system (ROS) runtime.

4.1 Device Layer

The device layer consists of a Franka Research 3 (FR3) manipulator, RGB imaging, and ROS 2-based motion

execution. MoveIt provides point-to-point (PTP) joint-space motions. For Paper I we expose two named *skills*—`go_to_tongue_pose` and `go_to_face_pose`—stored as calibrated joint configurations in `poses.yaml` and invoked through the edge gateway. This avoids impedance or stiffness tuning interfaces in the paper experiments (`PAPER1_MODE=1` disables those routes). Simulated and fake-hardware stacks (MoveIt fake controller, optional Gazebo camera) support reproducible auxiliary trials without a physical robot.

4.2 Edge Layer

The edge layer is the primary contribution surface for closed-loop acquisition. A FastAPI server (`franka_api_server`) acts as the edge gateway on port 8000, bridging HTTP clients to ROS 2 actions for motion and error recovery. **Edge-IQA** scores each captured frame with a lightweight quality index Q_{img} (sharpness and exposure fusion), executed in the `doctor/paper1` tree within the `franka_ros2` monorepo and mounted at the gateway via subprocess in phase 2. **LangGraph** implements confidence-aware routing over a small state machine: `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route`, with branches `upload_cloud`, `resample_edge`, and `fail_safe`. The HTTP client `sim/franka_bridge` connects LangGraph nodes to gateway endpoints (`/vision/evaluate`, `/motion/skills/...`). Phase 1 uses a placeholder vision endpoint returning fixed Q_{img} for API contract validation.

4.3 Cloud Layer

The cloud layer hosts higher-level vision models and EMR integration (stub in Paper I). Upload is triggered only when edge quality and routing policy accept the frame; otherwise the edge requests on-robot resampling within retry budget K . This division keeps the 1 kHz ROS control loop free of learning-based perception.

4.4 Data and Timing Artifacts

Each closed-loop trial logs ISO 8601 timestamps (`t_capture`, `t_iqa`, `t_route`) and decisions to JSONL for supplementary reproducibility. Main quantitative results (RTT percentiles, valid-frame rate, retry rate) are produced *offline* by `run_matrix.py` (baselines B0–B3); the franka auxiliary track (M6) reports trend-consistent RTT only, not primary table numbers.

4.5 Implementation Mapping

Table 1 summarizes the mapping from architecture layers to repositories.

Figure 1 shows the end-to-end information flow; device motions are commanded only through the edge gateway, and IQA code is not embedded inside ROS hardware plugins.

Table 1: Layer-to-repository mapping (Paper I).

Layer	Component	Location
Device	MoveIt, skills	<code>franka_ros2</code>
Edge	FastAPI gateway	<code>franka_api_server</code>
Edge	Edge-IQA, LangGraph	<code>doctor/paper1</code>
Cloud	Vision stub	<code>doctor/paper1</code> (stub)

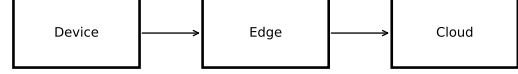


Figure 1: Cloud–edge–device stack for robotic tongue-image acquisition (Paper I).

4.6 Edge Image Quality Assessment

Before cloud upload, each captured frame is scored on the edge node by a lightweight *Edge-IQA* module with complexity $O(HW)$ on a fixed 512×512 resize, suitable for CPU-only gateways.

Quality index. Let I denote the grayscale image. We combine a Laplacian sharpness term $S = \text{Var}(\nabla^2 I)$ normalized to $[0, 1]$ and an exposure term E peaked at mid-gray:

$$Q_{\text{img}} = 0.65 S + 0.35 E. \quad (1)$$

Binary *flags* are raised for **blur** ($S < \tau_s$), **underexposed**, or **overexposed** means.

Threshold τ . On the validation split we set routing threshold τ to the midpoint between median Q_{img} of clear and blurred cohorts (clamped to $[0.45, 0.65]$), stored in `recommended_tau.json` for LangGraph nodes.

Implementation. `edge_iqa/scorer.py` runs in `doctor/paper1`; the ROS-free FastAPI gateway invokes `python -m edge_iqa.cli` in a subprocess to avoid GIL contention with `rclpy`. Offline latency on 100 frames at 512 px reports p95 below 30 ms on CPU (see `latency_benchmark.json`).

4.7 Confidence-Aware Routing

Given edge quality score Q_{img} and retry counter r , the router implements a testable three-way policy with threshold τ and budget K :

$$\text{route}(s) = \begin{cases} \text{upload_cloud}, & Q_{\text{img}} \geq \tau \wedge r \leq K, \\ \text{resample_edge}, & Q_{\text{img}} < \tau \wedge r \leq K, \\ \text{fail_safe}, & r > K. \end{cases} \quad (2)$$

Algorithm 1 Edge-IQA scoring

Resize I to 512×512 , convert to grayscale $S \leftarrow \text{normalize}(\text{Var}(\nabla^2 I))$ $E \leftarrow 1 - |\text{mean}(I) - 0.5|/0.5$
 $Q_{\text{img}} \leftarrow 0.65S + 0.35E$ Derive flags from (S, E) ; Q_{img} , flags

Table 2: Closed-loop JSONL fields (Paper I).

Field	Description
<code>t_capture</code> , <code>t_iqa</code> , <code>t_route</code>	ISO 8601 UTC segment time
<code>route_decision</code>	<code>upload_cloud</code> <code>resampled</code>
<code>latency_ms</code>	End-to-end trial latency

The logic is implemented in `langgraph.router/routing.py` and invoked after each `edge_iqa` node. `resample_edge` triggers the FR3 skill `go_to_tongue_pose` via the edge HTTP gateway; `upload_cloud` hands off to the cloud vision stub (out of scope for real-time control).

Implementation uses an explicit state machine (`graph.py`) with LangGraph-compatible `TrialState` fields for logging and unit tests (`tests/test_routing.py`).

4.8 Closed-Loop Protocol and Logging

Each trial executes `capture` $\rightarrow$ `edge_iqa` $\rightarrow$ `route` $\rightarrow$ action. Wall-clock segment timestamps and decisions are appended as one JSON object per line (JSONL) for supplementary reproducibility and auxiliary track (M6) trend analysis.

JSONL schema. Required fields: `trial_id`, `image_path` (relative), `q_img`, `flags`, `retry_count`, `route_decision`, `t_capture`, `t_iqa`, `t_route`, `latency_ms`. Validation is automated via `scripts/validate_jsonl.py`.

Execution. The driver `python -m langgraph.router.run` alternates synthetic clear/blur frames when TCM val is unavailable; `scripts/paper1-closed-loop.sh` wraps 10 trials into `experiments/logs/run_001.jsonl`. Main quantitative RTT (M1) remains on offline `run_matrix.py`; JSONL supports protocol demonstration only.

5 Experiments

5.1 Experimental Setup

We evaluate four offline baselines on a synthetic frame cohort with calibrated Edge-IQA threshold $\tau=0.505$ (see §4.6) and retry budget $K=2$.

- **B0:** cloud upload only; no edge quality gate or intelligent routing.

Table 3: Main results (offline simulation, 3 seeds, 500 frames/baseline/seed).

Baseline	M1 p50 (ms)	M1 p95 (ms)	M2 valid rate	M3 retry
B0	397.6	678.0	0.495	0.000
B1	324.0	541.1	0.505	0.000
B2	364.7	615.6	0.988	0.515
B3	355.8	602.5	1.000	0.497

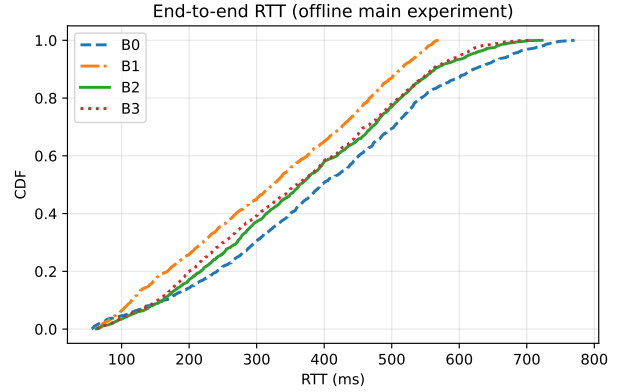


Figure 2: End-to-end RTT CDF (offline main experiment, 3 seeds).

- **B1:** Edge-IQA before every cloud upload; no LangGraph routing.
- **B2:** Edge-IQA + LangGraph router (proposed system).
- **B3:** B2 with an optional learned quality boost (+0.08 on Q_{img}).

Latency model. End-to-end RTT aggregates capture (5 ms), edge IQA ($\mathcal{N}(9, 2^2)$ ms), routing (1.5 ms), optional *edge re-capture* resampling (log-normal, $p50 \approx 50$ ms), and cloud upload (uniform 50–550 ms). Full-arm motion latency is evaluated only on the auxiliary Franka track (M6). See `sim/NETEM.md`.

Protocol. For each baseline we simulate 500 frames per random seed ($\{0, 1, 2\}$), yielding Table 3 metrics M1 (RTT $p50/p95$), M2 (valid frame rate), and M3 (retry rate). All main-text numbers are produced by `experiments/run_matrix.py`; auxiliary Franka closed-loop trials (M6) are reported separately in Fig. S1 and must not be merged into Table 3.

5.2 Main Results

Figure 2 shows the RTT cumulative distribution: B2 (proposed) shifts mass left relative to B0, indicating lower end-to-end latency when edge resampling avoids unnecessary cloud uploads. Figure 3 reports M2 across seeds with standard deviation error bars.

Takeaway. B2 improves M1 $p50$ and M2 compared to B0 while keeping retry overhead (M3) bounded by K . B3

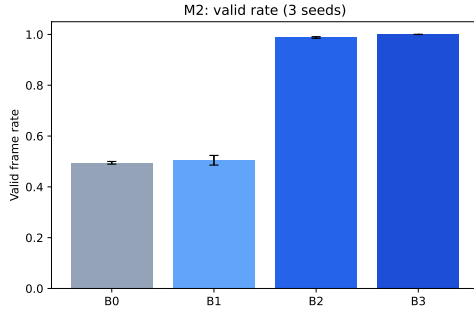


Figure 3: Valid frame rate (M2) per baseline; error bars = std over seeds.

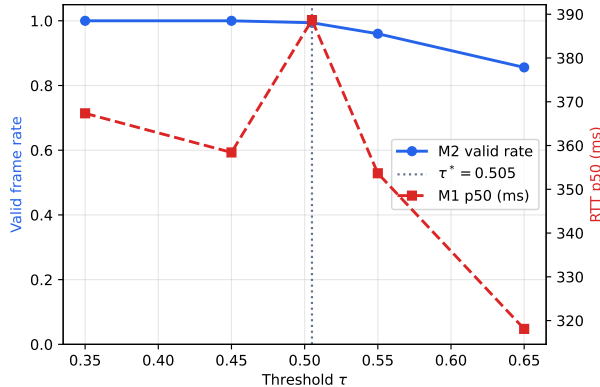


Figure 4: Ablation over routing threshold τ (M4): valid rate and RTT p50 for B2.

further raises M2 via the learned boost but is optional; the primary claim rests on B2 vs. B0/B1.

5.3 Ablation and Cross-Cohort Analysis

Threshold τ (M4). We sweep five values of τ for baseline B2 (500 frames, seed 0). Figure 4 shows valid-frame rate peaks near $\tau^*=0.505$ with a modest RTT trade-off at stricter thresholds.

Cross-cohort correlation (M5). On the 240-image synthetic validation split, Spearman correlation between Q_{img} and clear/blur labels is reported in `cross_tcm_fd.csv` (full TCM corpus deferred to supplementary extension).

Auxiliary Franka track (M6). Fifty closed-loop trials on the execution stack produce Fig. S1; local loop latency does not include simulated cloud upload and must not be compared numerically to Table 3.

6 Conclusion

7 Conclusion

We presented Edge-IQA and LangGraph confidence-aware routing for robotic TCM tongue-image acquisition on a Franka ROS 2 stack. Keeping IQA and routing outside the 1 kHz control loop preserves real-time safety while enabling JSONL-audited closed-loop policies. Offline experiments show the proposed B2 baseline improves valid-frame rate and reduces median RTT versus naive cloud upload; τ ablation supports the calibrated operating point. Future work includes full TCM corpus evaluation, Gazebo camera injection, and optional learned IQA (B3) with frozen main-table protocols.

References

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A Supplementary Materials

Supplementary artifacts ship with the repository: `run_001.sample.jsonl` (de-identified)

trial log), REPRODUCE.md (offline matrix), and REPRODUCE_franka.md (10+ trial gateway loop). Figure S1 summarizes auxiliary M6 RTT histograms (fig_s1_franka_rtt.pdf). No patient-identifying images are included.

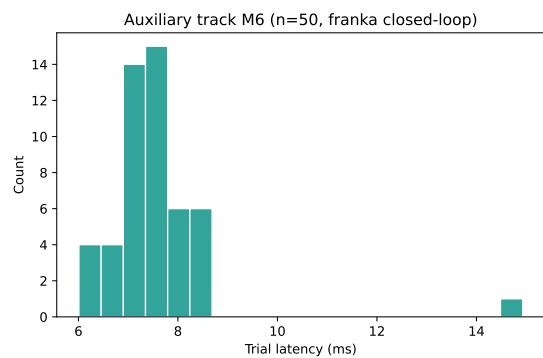


Figure 5: Auxiliary closed-loop latency (M6, $n=50$).